

Vericoding

Verification-Driven Development

From behavioural features to machine-checked proofs

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Vibecoding

The status quo of LLM-driven development.

You prompt. The model emits. It runs.

What you leave behind:

- No specification of intended behaviour
- No evidence of correctness
- No conformance regime for the next generation

The only durable artifact is the code itself.

Vericoding

The methodology specsaver operationalises.

You write a specification. The spec survives.

What you leave behind:

- Features — Gherkin scenarios domain experts can review
- Contracts — mathematical pre/post-conditions
- Proof obligations — machine-checked in Rocq/Coq

Generated code is a replaceable implementation detail.

Two Development Flows

Retrospective

- Feature — Gherkin scenario tables
- Implementation — write the production code
- Testing (sanity) — make sure it basically works
- Contract — capture observed behaviour as mathematical spec
- Testing (dialectic) — re-run under contract checking; either contract or implementation is wrong — the dialectic
- Formal Proof — lower to Coq, LLM closes obligations; DISPROVED → back to dialectic

Contract-First

- Feature — Gherkin scenario tables
- Contract — write spec as acceptance criteria
- Implementation — build code against the specification
- Testing — validate that implementation satisfies contract
- Formal Proof — machine-checked against kernel

Both converge: the contract is the lasting artifact.
DISPROVED yields witnesses → back to the dialectic.

Contract Language

The pure terminating fragment
of Python

Every clause is a boolean-valued lambda:

- `requires(state, args)` — admissibility
- `ensures(state, args, result, new_state)` — success post
- `when(state, args)` — exception conditions
- `invariant(state)` — global legality

Static purity check rejects side effects, loops, and non-boolean returns before execution.

Anatomised as mathematical
record

$$C = \langle \Sigma, \text{args}, \text{pre}, \text{post}, \\ \mathcal{X}, \mathcal{I}, \mathcal{D}, \mathcal{W}, \mathcal{R}, \mathcal{G} \rangle$$

$\mathcal{X}$ — exceptional exits (when + ensures + frame)

$\mathcal{I}$ — invariants on every state

$\mathcal{D}$ — derived-state definitions

$\mathcal{W} / \mathcal{R}$ — semantic frame

$\mathcal{G}$ — ghost state

Semantic Frames

Declare what you write, not what you don't.

Write paths

```
writes = {  
  "state.products[sku].reserved",  
  "state.invitations[token]",  
  "state.reservation_log",  
}
```

Derived obligations



Every other keyed attribute unchanged



Every other keyed row untouched



All other event channels silent



Deletions always rejected



Keyed-row inserts allowed for args-resolved keys

The checker derives frame soundness from write paths alone. No "everything else unchanged" clauses needed.

Exception Exits

Exceptions are data, not control flow.

```
exceptions = [  
  ExcExit(  
    raises = InsufficientStockError,  
    when   = [lambda s, a:  
                s.products[a.sku].on_hand  
                - s.products[a.sku].reserved < a.quantity],  
    writes = {"state.failure_log"},  
    ensures = [lambda s, a, e, s':  
                extends_by_one(  
                  s.failure_log, s'.failure_log,  
                  λf. f.sku == a.sku  
                    ∧ f.reason == e.code)]  
  )  
]
```

The runner checks: correct exception raised, failure telemetry contains exactly one new event with exact fields, nothing outside the exit's writes changed.

Symmetric Projection

Symmetric Projection — Runner

1. materialize

witness → temp SQLite DB + engine + EventLog

2. pre-check

invariants on pre-state; admissibility (requires)

3. execute

service runs against real SQLite via SQLAlchemy;
wrapper emits typed events into the log

4. frame check

everything outside writes unchanged

5. derived check

derived fields $\equiv$ recomputed from observed

6. post-check

ensures (or exit ensures) for the outcome;
invariants on post-state

Domain Declaration

One object → runners, CLI, conformance suite.

```
inventory = SqlDomain(  
    name          = "inventory",  
    package       = "examples.inventory",  
    materializer  = SqlMaterializer(ddl, TableSpec[]),  
    projection    = SqlProjection(types, hooks),  
    operations    = [  
        SqlOperation(contract, wrapper,  
                        witness_builder, feature_file),  
        ...  
    ]  
)
```

scenario runners

CLI dispatch

conformance tests

Adding a domain: types + service + contracts + features + witness builders + one declaration. Everything else (materializer, projection, runner wiring, test dispatch) is derived.

Theory Stack

Services run unmodified on differentially-tested stubs.

Service ordinary SQLAlchemy / logging / OTel API code

Shim make_engine, make_log_capture, make_otel_capture

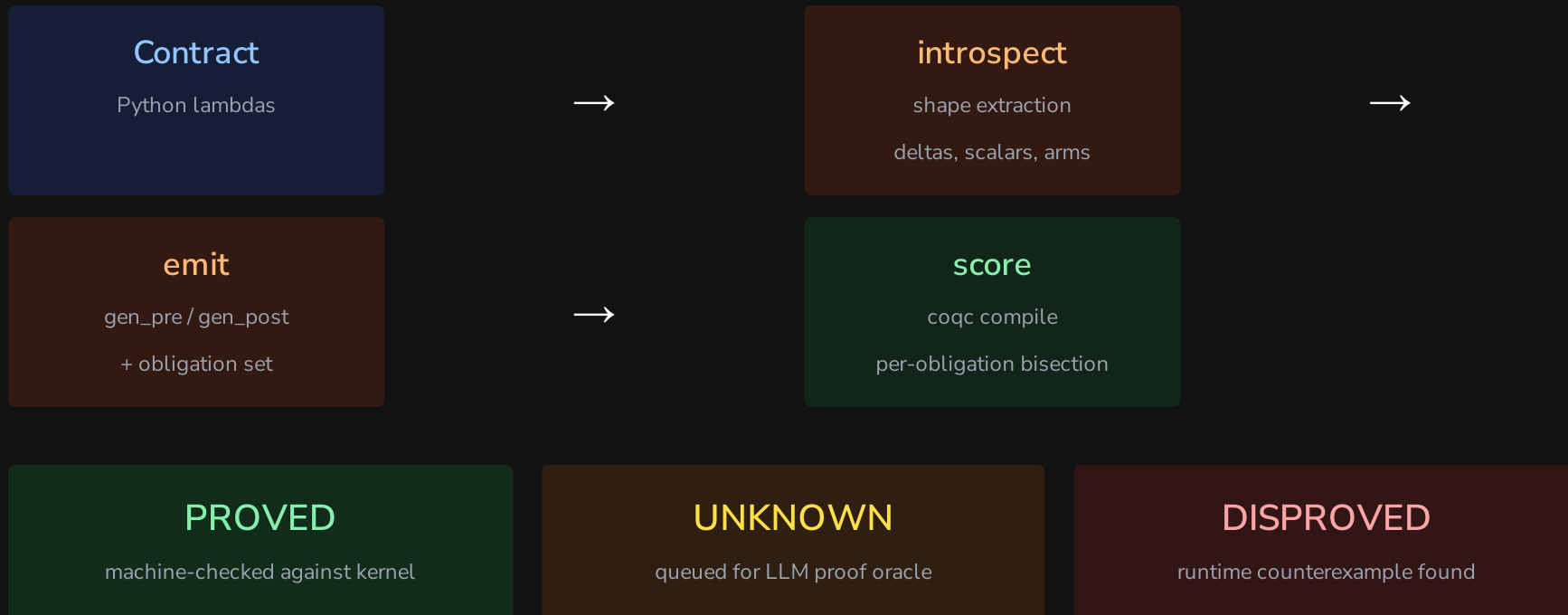
StubHandler operational semantics over the state model

State model TableStore / LogStore / OtelStore

Trace final interpretation (events)

Theory conformance: Stub and real library agree observationally on a differential suite; translator rejects out-of-fragment input loudly; state model and trace are pure data.

Lowering Pipeline



Example: Inventory Reserve

Pre-condition:

$$\text{pre}(s, a) = a.\text{quantity} > 0 \wedge a.\text{sku} \in s.\text{products}$$

Post-condition (excerpt):

$$\begin{aligned} \text{post}(s, a, r, s') &= s'.\text{products}[a.\text{sku}].\text{reserved} = s[\cdot].\text{reserved} + a.\text{quantity} \\ &\wedge \text{extends_by_one}(s.\text{reservation_log}, s'.\text{reservation_log}, \lambda e. e.\text{sku} = a.\text{sku}) \\ &\wedge \text{implies}(\text{crossed}(s, a, s'), \text{extends_by_one}(s.\text{alert_log}, s'.\text{alert_log}, \dots)) \end{aligned}$$

Exception exit:

$$\mathcal{X} = \{ \langle \text{InsufficientStock}, s.\text{available}(a.\text{sku}) < a.\text{quantity}, \{ \text{failure_log} \}, \text{failure_ensures} \rangle \}$$

Example: Invitations

Invite (insert frame):

$$\begin{aligned} \text{post}_{\text{invite}}(s, a, r, s') &= s'.\text{invitations}[a.\text{token}].\text{status} = \text{pending} \\ &\quad \wedge s'.\text{invitations}[a.\text{token}].\text{expires_at} = a.\text{now} + 7 \cdot 86400 \end{aligned}$$

Accept (multi-table delta):

$$\begin{aligned} \text{post}_{\text{accept}}(s, a, r, s') &= s'.\text{invitations}[a.\text{token}].\text{status} = \text{accepted} \\ &\quad \wedge s'.\text{members}[a.\text{user_id}].\text{org_id} = s.\text{invitations}[a.\text{token}].\text{org_id} \\ &\quad \wedge s'.\text{users}[a.\text{user_id}].\text{default_org} = s.\text{invitations}[a.\text{token}].\text{org_id} \end{aligned}$$

Design decisions for verifiability: Token is a caller-supplied argument (args-resolved insert key). **now** is an integer epoch argument (expiry is a pure comparison).

